Vergabepilot.AI

Review Party#2 · Progress Update

Agentic AI for Procurement Document Extraction

June 2026

· CORE Research Group × Ciconia Systems GmbH

RECAP

The Problem & Our Solution

The Problem

- Every new procurement portal needs a scraper built by hand.
- Thousands of fragmented German & EU tender portals.
- Engineering effort becomes the bottleneck — slow, costly, hard to scale.

Our Solution

- An agentic cascade pipeline that auto-generates, validates and runs scrapers.
- Degrades from cheap cached strategies to autonomous visual agents.
- New portals are onboarded with little to no manual code.

ARCHITECTURE

How the System Is Built

Next.js UI
Dashboard

FastAPI
/api/jobs

Celery Workers
async pool

Redis
broker

Phase 3 — Cascade

- pipeline.py router
- modular, sequential

Phase 1 — Generation

- generator + validator
- sandbox executor + evaluator

Phase 2 — CUA

- Playwright orchestrator
- visual fallback agent

MinIO (S3)
persistent documents

PostgreSQL
job & run metadata

Fail-safe, cost-aware orchestration — each request flows through the cheapest viable strategy first.

STRATEGY

The Cascade — Cheapest Viable Path First

1. **Manual Scraper** — Pre-written, high-reliability scripts for top domains
2. **Existing Scraper** — Auto-saved from previous successful LLM runs
3. **Deterministic Template** — Direct ZIP download for DTVP / Satellite portals
4. **LLM-Generated Scraper** — Synthesised on-the-fly inside PySandbox
5. **Computer-Use Agent** — Visual fallback — screenshot → action
6. **Failure / None** — All strategies exhausted

cost · capability · latency

PHASE 1

LLM Scraper Generation & Feedback Loop

Generator

Takes a URL and synthesises a Python / Playwright scraper via an LLM, per domain.

Validator

Checks generated code for safety using AST analysis before any execution.

Executor

Runs scrapers in a sandboxed subprocess with strict memory & timeout limits.

Evaluator

Compares results against expected outcomes and flags discrepancies.

Self-healing loop: up to 3 retries, repairing the code from stdout / stderr logs before falling back to the CUA.

PHASE 2

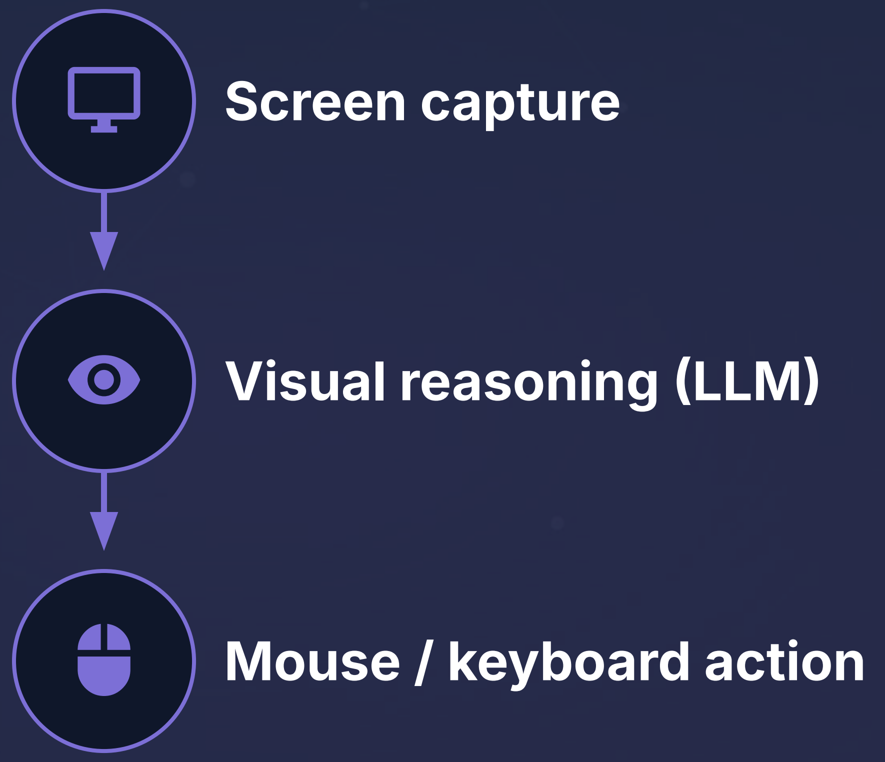
Computer-Use Agent — Visual Fallback

Fail-Safe Mechanism: The ultimate fallback when generated code fails to execute or locate elements.

Human-Like Interaction: Captures real-time screenshots and reasons over the visual UI like a user would.

Direct Action: Acts via mouse coordinates and keyboard input directly on the portal interface.

Anti-Scraping Shield: Bypasses modern security paywalls that successfully block headless scripts.



RESULTS

Scraper Generation — Production Run

97%
success on 100 live URLs

$0.02
LLM cost per cached run

348s
total run time (4 workers)

75 / 4
cached vs new LLM calls

How the cache pays off

Production (cached registry): **97 / 100 success** · only 4 LLM retries · **~$0.02**

Fresh full re-generation: **87 / 100** · 79 + 18 retries · **~$0.53**

Multi-Model Benchmarking

Rank	Model	Success	Latency	Cost / run	Grade
#1	Gemini 2.5 Flash Lite	83%	4,176 ms	$0.00181	A
#2	**Gemini 2.5 Flash**	83%	4,540 ms	$0.00338	**A**
#3	**GPT-4o mini**	67%	12,786 ms	$0.00273	**B**
#4	**Claude Sonnet**	67%	17,705 ms	$0.07191	**B**
#5	**Gemini 2.5 Pro**	50%	24,989 ms	$0.05495	**D**

Key takeaways

- Gemini 2.5 Flash Lite wins — best accuracy at the lowest cost & latency.
- Expensive frontier models underperformed on this task.
- We default to the cheap, fast model and escalate only when needed.

Project Status & Roadmap

CURRENT

Discovery	Phase 1	Phase 2	Phase 3	Final Phase
Research & insights	Manual scraper + benchmarking	CUA agent live + analysis	Cascade pipeline + error injection	Frontend ↔ backend, server-side loading

Four of five phases complete — the full integrated system & live demo land in the Final Presentation.

NEXT

Future-Plan

Connect frontend ↔ backend

Wire the Next.js dashboard to live FastAPI jobs with server-side loading.

End-to-end live demo

Submit a URL and watch the cascade pick a strategy in real time.

Hardening & error injection

Stress-test the pipeline and the security / diagnostics monitor.

Final metrics & polish

Refresh benchmarks at scale and finalise the admin command center.

Member

Team Members

Thank You

Questions & discussion welcome

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